SideQuest:
Gamified Local Community & Gig Platform

Fundamentals of Software Engineering

Deliverable Number: Deliverable # 2 (Iteration 1 / Sprint 1)

Date of Submission: March 24, 2026

Group Members:

- Talmeez ur Rehman (Roll Number: 24L-2576) - Project Lead & Frontend UI/UX Developer
- Muhammad Yaseen (Roll Number: 24L-2515) - Backend Developer & Software Architect
- Taimur Amir (Roll Number: 24L-2518) - Requirement Engineer & QA Tester

Project Title and Introduction:

Project Title: SideQuest: Gamified Local Community & Gig Platform

Introduction: SideQuest is a hybrid community-building and gig-economy web application designed to gamify daily tasks and social needs into localized "Quests." Sprint 1 focuses on the core module: **Core Setup, Authentication, and Quest Board**. This initial iteration establishes the foundational database architecture, secure user registration and login flows, and the primary frontend interfaces required for users to view, filter, and post localized tasks.

User Stories (Sprint Backlog):

US-01: The Quest Board

Story ID	US-01	**Story Title**	The Quest Board
Priority	High	**Estimated Hours**	8
User Story	**As an** adventurer, **I want** to view and filter a board of available quests **so that** I can easily find tasks or social events happening around me.		
Description	This story involves developing the main dashboard (Notice Board) where all active quests are displayed in a grid format. It includes filtering capabilities to allow users to sort between "Errands", "Social", and "All" quests.		
Sub User Stories	• As a user, I want to view a list of all active quests with their title, description, reward, and location. • As a user, I want to filter the board to only show "Social" or "Errand" quests.		
Acceptance Criteria	• The system must fetch quests from the database on load. • Quests must be displayed as individual cards. • A filter mechanism must be present and update the UI without reloading the page.		

US-02: Posting a Bounty

Story ID	US-02	**Story Title**	Posting a Bounty
Priority	High	**Estimated Hours**	6
User Story	**As a** quest-giver, **I want** to create and post new quests **so that** other adventurers can see them and offer their help.		
Description	This story covers the creation of a modal form that allows users to submit a new quest to the database. It includes logic for quick-cast templates and conditional logic for specific quest types (e.g., locking the reward field for social events).		
Sub User Stories	• As a user, I want to fill out a form with title, type, reward, and location to post a quest. • As a user, I want to click quick-templates to automatically fill out common quests. • As a system, I want to prevent users from adding a monetary reward to "Social" quests. • As a user, I want to add expiry time for my quests.		
Acceptance Criteria	• Form must include validation for required fields. • Selecting "Social" must automatically set the reward to 0 and disable the input. • Upon successful posting, the board must refresh to show the new quest.		

US-03: Bookmarking Quests

Story ID	US-03	**Story Title**	Bookmarking Quests
Priority	Medium	**Estimated Hours**	4
User Story	**As a** user, **I want** to save or bookmark specific quests **so that** I can easily find and track them later without searching the whole board.		
Description	This story involves creating a many-to-many relationship in the database for saved quests, allowing users to toggle a bookmark icon on quest cards, and providing a dedicated view for their		

	saved items.
Sub User Stories	• As a user, I want to click a bookmark icon on a quest to save it. • As a user, I want to view only my saved quests using a dedicated filter tab.
Acceptance Criteria	• Clicking the bookmark icon must toggle its visual state (filled/unfilled). • Saved quests must persist in the database linked to the user's ID. • The "Saved" filter tab must exclusively display bookmarked quests.

US-04: Guild Authentication

Story ID	US-04	**Story Title**	Guild Authentication
Priority	Critical	**Estimated Hours**	8
User Story	**As a** new user, **I want** to register and log in securely using my CNIC **so that** the platform maintains a safe, verified community of local users.		
Description	This story covers the backend and frontend implementation of JWT-based authentication. It includes secure password hashing (bcrypt), CNIC validation, and protected routes to ensure only logged-in users access the app.		
Sub User Stories	• As a user, I want to register an account with my Name, Email, CNIC, and Password. • As a user, I want my CNIC input to auto-format with hyphens as I type. • As a user, I want to log in and be redirected to the main quest board.		
Acceptance Criteria	• CNIC must be strictly validated on the frontend (XXXXX-XXXXXXX-X). • Passwords must be hashed in the database. • Unauthenticated users must be forced to the login screen.		

Structured Specifications

US-01.1: View and fetch quests

Type	Specification
Functional	The application must fetch all quests with a status of 'Open' from the backend upon initial render.
Functional	The Quest Board must map incoming data to individual UI cards displaying Title, Type, Reward, Location, and Creator.
UX	A loading state (e.g., "Consulting the Oracles...") must be displayed while the API request is pending.

US-01.2: Filter by type

Type	Specification
Functional	The filter controls must pass a type query parameter (e.g., ?type=Social) to the backend API.
UX	The currently active filter tab must be visually highlighted (distinct background/border).
Performance	The board must update the displayed quests without triggering a full page reload.

US-02.1: Create custom quest

Type	Specification
Functional	The submission form must require Title, Quest Type, Details, and Location to be filled before allowing submission.
Functional	The backend must extract the QuestGiverID securely from the JWT token, not from the client request body.

UX	The form must appear as a modal overlay, darkening the background to focus user attention.

US-02.2: Quick-cast templates & Conditional Logic

Type	Specification
Functional	Clicking a template button must immediately populate the form state with predefined title, type, and reward data.
Functional	If "Social" is selected as the Quest Type, the system must force the Reward state to 0.
UX	When "Social" is selected, the Reward input field must be disabled and visually greyed out to indicate it is locked.

US-03.1: Toggle bookmark state

Type	Specification
Functional	The system must check the SavedQuests table; if a record exists, the toggle removes it, otherwise it inserts a new record.
UX	The bookmark icon must instantly change to a "filled" state on click, using optimistic UI updates before the server responds.

US-04.1: Register with CNIC format

Type	Specification
Functional	The input field must strip all non-numeric characters and automatically insert hyphens after the 5th and 12th digits.
Functional	The form must prevent submission and show an error if the CNIC does not strictly match the 13-digit pattern.

UX	Error messages must be displayed clearly in a high-contrast box (e.g., red background) above the form.

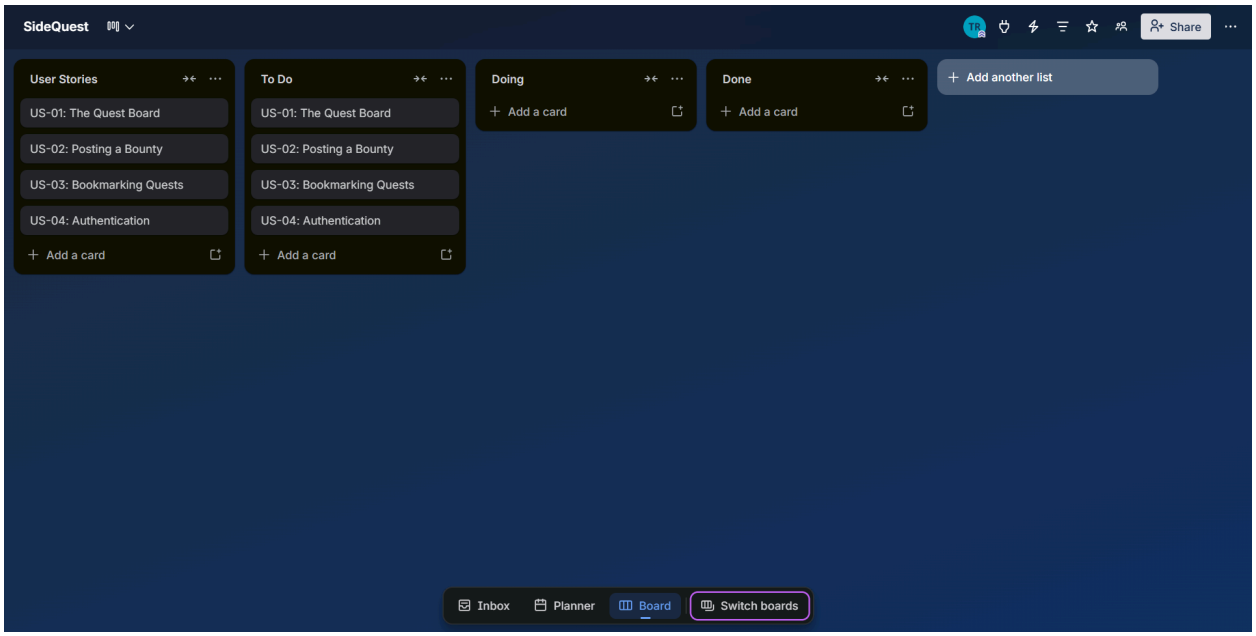
US-04.2: Login securely

Type	Specification
Functional	The backend must verify the hashed password using bcrypt and issue a JSON Web Token (JWT) valid for a set duration.
Functional	The client application must store the JWT in localStorage and attach it to the Authorization header of all subsequent API requests.
Security	Private routes (like the Quest Board) must redirect to /login if no valid token is found in the application state.

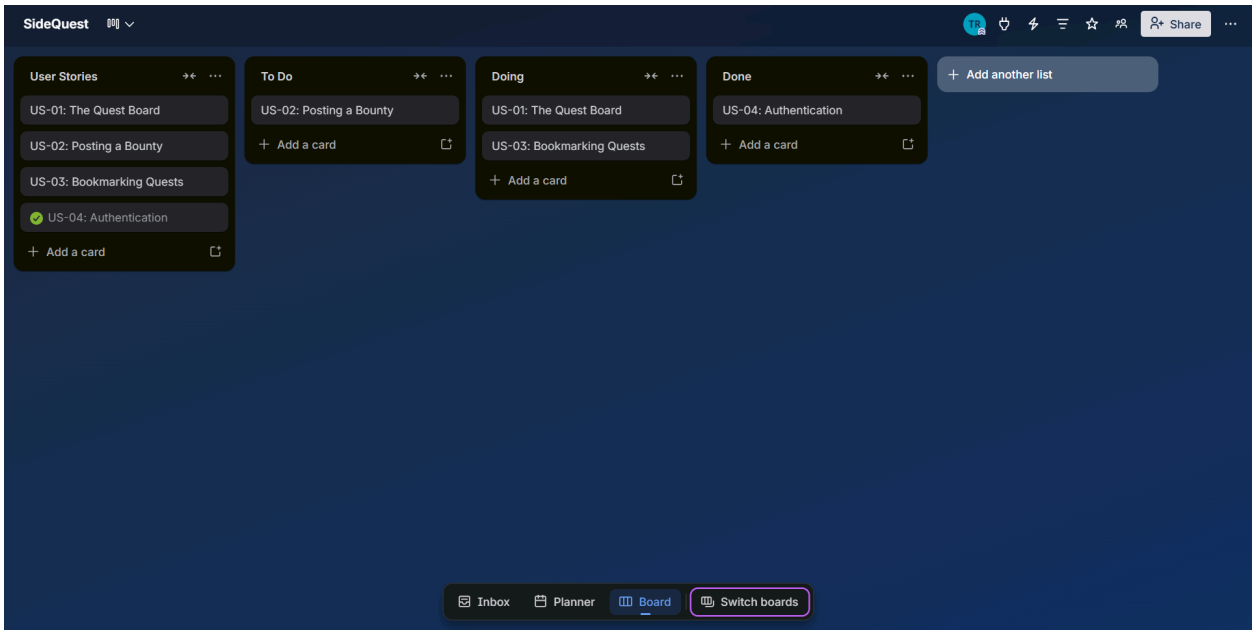
Scrum Board

(Trello board link: https://trello.com/invite/b/69bda26c79ab58133808b311/ATTI3b3a5945ffd93cb03da3050e104f2e035DD1600A/sidequest)

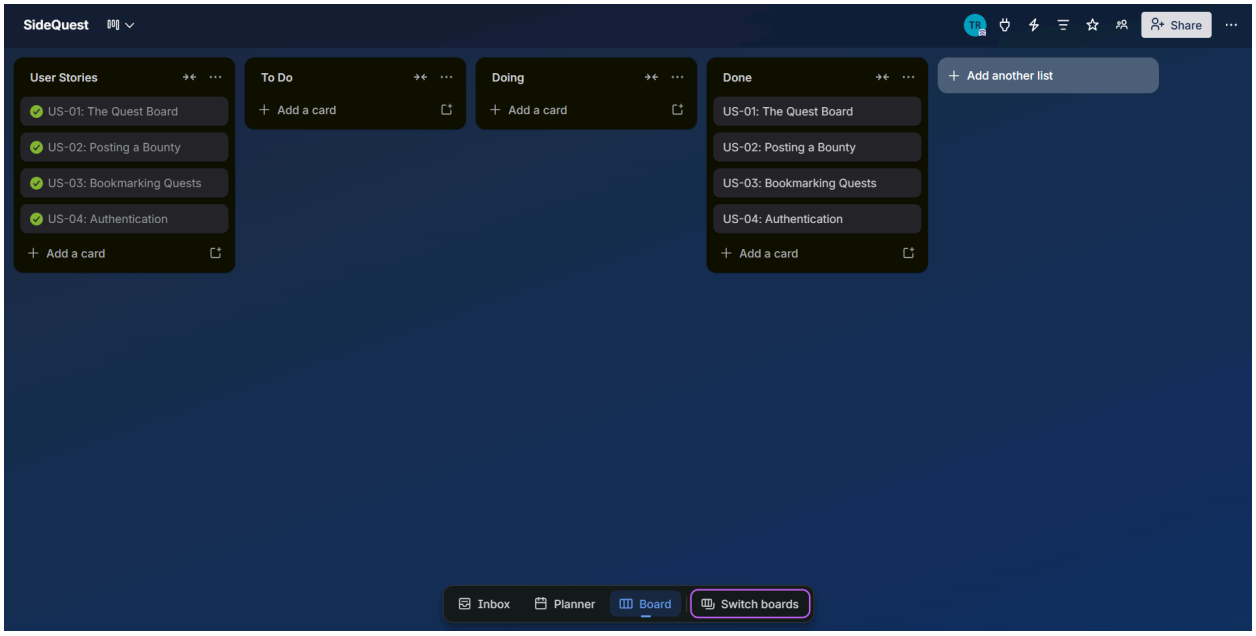
- **Snapshot 1 (Start of Sprint):**



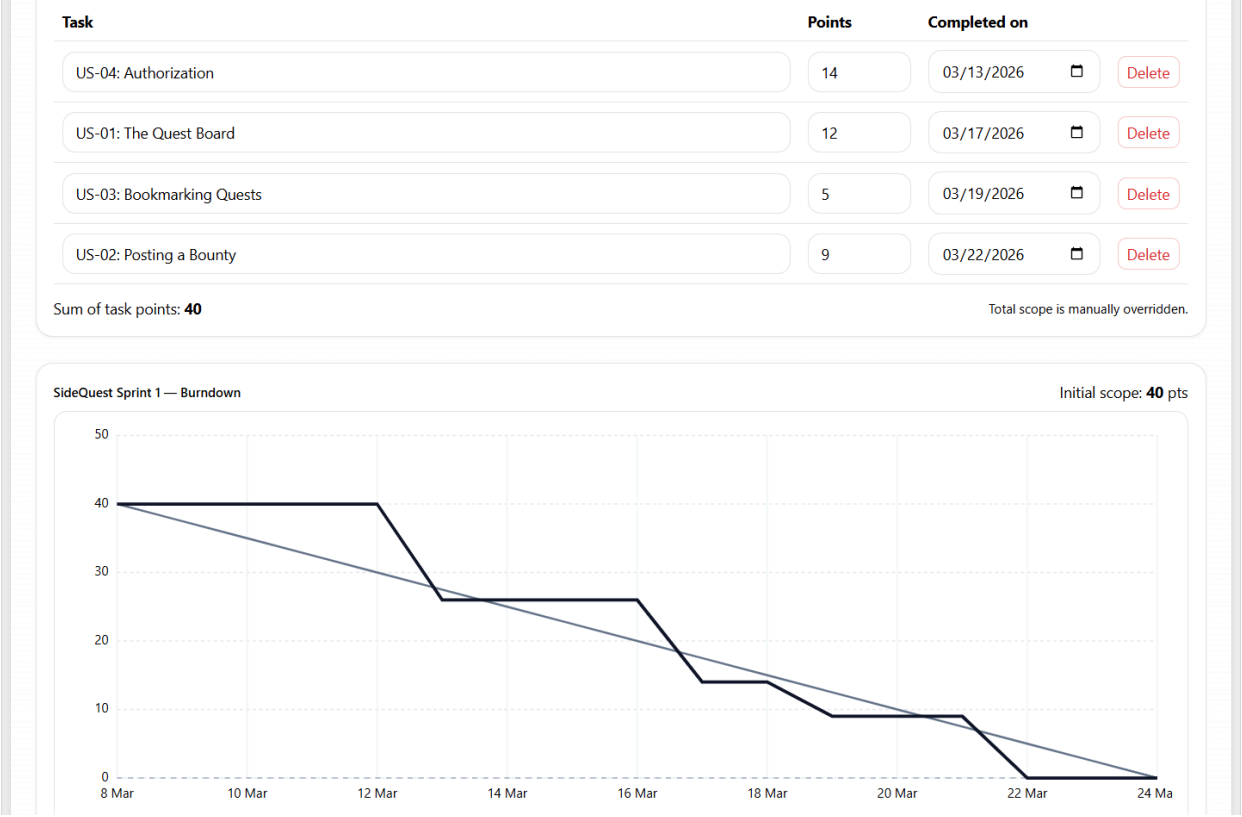
- **Snapshot 2 (Middle of Sprint):**



- **Snapshot 3 (End of Sprint):**



- **Product Burn-Down Chart:**



Task	Points	Completed on	
US-04: Authorization	14	03/13/2026	Delete
US-01: The Quest Board	12	03/17/2026	Delete
US-03: Bookmarking Quests	5	03/19/2026	Delete
US-02: Posting a Bounty	9	03/22/2026	Delete

Sum of task points: **40**

Total scope is manually overridden.

NFR Specification:

- **Security:** All user passwords must be hashed using industry-standard bcrypt before database insertion. Application endpoints must be protected using JSON Web Tokens (JWT) to prevent unauthorized access.
- **Performance:** The platform must be highly responsive. The API endpoint responsible for fetching the Quest Board should return a payload of up to 50 quests in under 500 milliseconds under normal server load.
- **Usability:** The web application must employ responsive web design principles (via Tailwind CSS) to ensure all interfaces, forms, and quest cards are fully operable and aesthetically pleasing on both desktop monitors and mobile device screens.

Iteration 1 Implementation Screenshots

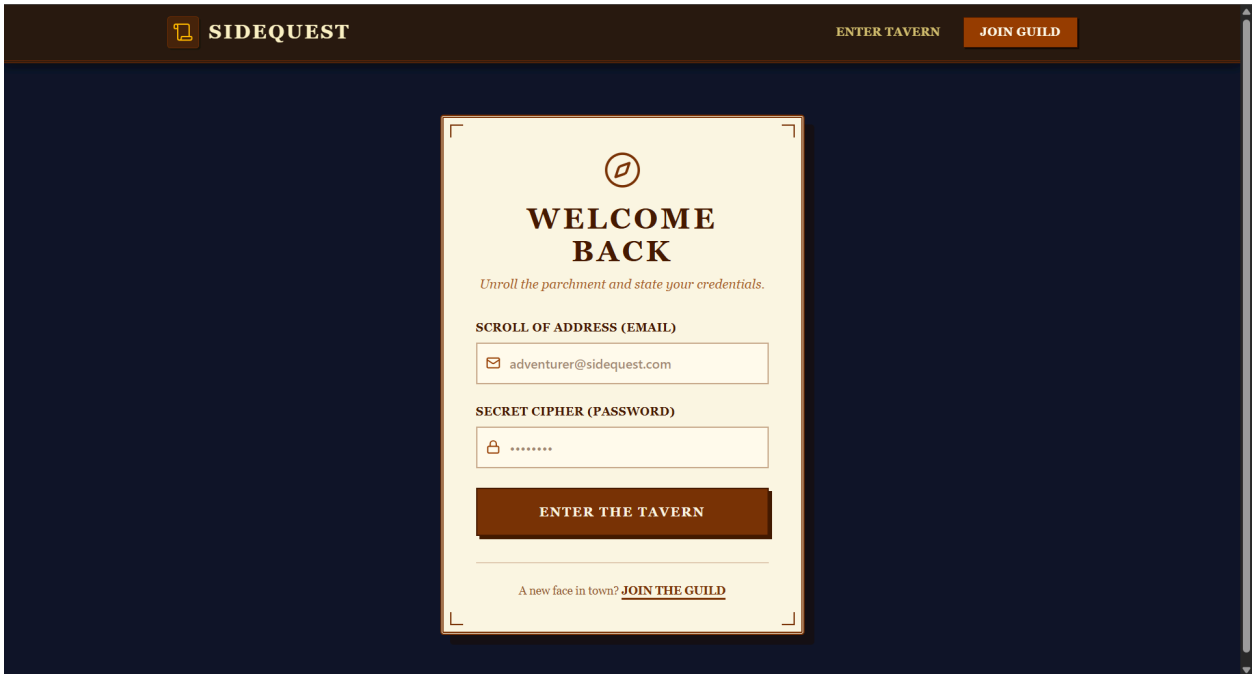
- **Database Tables (SQL Server):**

	UserID	FullName	Email	PasswordHash	CnicNumber	IsKycVerified	Rating	CreatedAt
1	6	Test User	test@sidequest.com	$2b$10$mMPmFGwnuwzCl0FfSQ4yW.ne/112.d6f6gJHzjUeU...	12345-6789012-3	0	0.00	2026-03-18 03:48:14.213
2	7	Talmeez	talmeez450@gmail.com	$2b$10$xgBvk8qbegAcSDKPNNY4He521Z.J0.L6gfmfHyroCu...	36103-8958896-1	0	0.00	2026-03-19 16:29:37.287
3	8	Gilgamesh	archer@fate.com	$2b$10$XNZfhb0a15QhAyFbCenHu.e441BuzTh1BbE88Cinza...	12333-2233312-1	0	0.00	2026-03-19 19:13:52.020

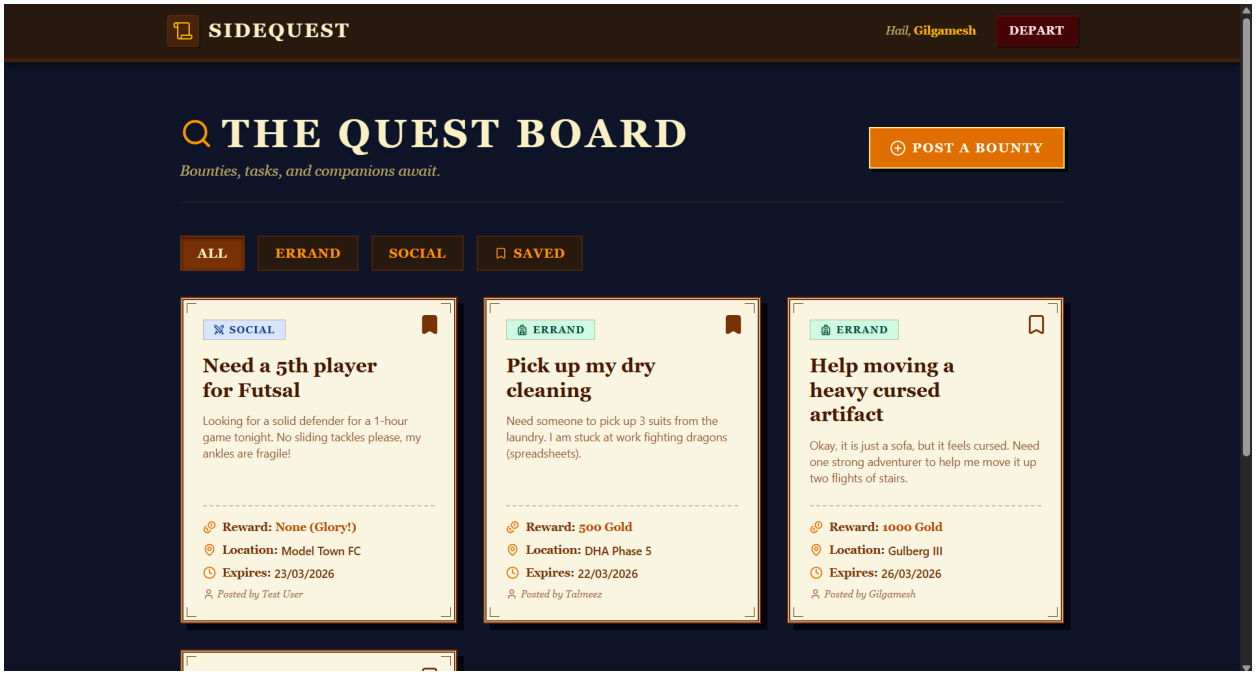
	QuestID	QuestGiverID	Title	Description	QuestType	Status	RewardAmount	Location	CreatedAt	ExpiresAt
1	2	6	Need a 5th player for Futsal	Looking for a solid defender for a 1-hour game toni...	Social	Open	0.00	Model Town FC	2026-03-20 23:54:31.040	2026-03-22 23:54:31.040
2	3	7	Pick up my dry cleaning	Need someone to pick up 3 suits from the laundry. ...	Errand	Open	500.00	DHA Phase 5	2026-03-20 23:54:31.040	2026-03-21 23:54:31.040
3	4	8	Help moving a heavy cursed artifact	Okay, it is just a sofa, but it feels cursed. Need one ...	Errand	Open	1000.00	Gulberg III	2026-03-20 23:54:31.040	2026-03-25 23:54:31.040
4	5	6	Board Game Night Tavern Gather	Hosting a D&D one-shot. Snacks provided, bring y...	Social	Open	0.00	Johar Town	2026-03-20 23:54:31.040	2026-03-27 23:54:31.040

	UserID	QuestID	SavedAt
1	6	4	2026-03-20 23:55:45.823
2	7	2	2026-03-20 23:55:45.823
3	8	2	2026-03-20 23:57:40.870
4	8	3	2026-03-20 23:55:45.823

- **Frontend - Login/Registration UI:**


SIDEQUEST
ENTER TAVERN
JOIN GUILD
WELCOME BACK
Unroll the parchment and state your credentials.
SCROLL OF ADDRESS (EMAIL)
adventurer@sidequest.com
SECRET CIPHER (PASSWORD)
••••••••
ENTER THE TAVERN
A new face in town? JOIN THE GUILD

- **Frontend - Quest Board UI:**



Work Division

- **Talmeez ur Rehman (Project Lead):** Frontend UI/UX design setup, React framework initialization, Tailwind CSS components, and frontend API integration.
- **Muhammad Yaseen:** Express.js server initialization, SQL Server database schema creation, andAPI endpoint development for Authentication and Quest management.
- **Taimur Amir:** Requirement engineering, structured specification documentation, Scrum/Trello board management, authorization form validation and manual API endpoint testing with Postman.